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CVE XXXX-XXXX

CVE-XXXX-XXXX 



Medium Binary Exploitation picoCTF 2022

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Hints ?

Description

1

Enter the CVE of the vulnerability as the flag with the correct flag format:

`picoCTF{CVE-XXXX-XXXXX}` replacing XXXX-XXXXX with the numbers for the matching vulnerability.

The CVE we're looking for is the first recorded remote code execution (RCE) vulnerability in 2021 in the Windows Print Spooler Service, which is available across desktop and server versions of Windows operating systems. The service is used to manage printers and print servers.

20,991 users solved

 49% Liked 

 `picoCTF{FLAG}`

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: CVE XXXX-XXXX

Category: Binary Exploitation

Date: 03/31/26



I. Objective

The objective of this challenge is to understand how to **identify real-world vulnerabilities** using public vulnerability databases.

The goal is to research a known vulnerability related to the Windows Print Spooler service and determine its **CVE identifier**, then submit it as the flag.

II. Background

In cybersecurity, vulnerabilities are publicly tracked using a system called **CVE (Common Vulnerabilities and Exposures)**.

Each vulnerability is assigned a unique ID so that:

- researchers can track it
- developers can fix it
- security professionals can understand the risk

In this challenge, the vulnerability is related to the **Windows Print Spooler service**, which had a serious flaw in 2021 that allowed attackers to execute malicious code remotely.

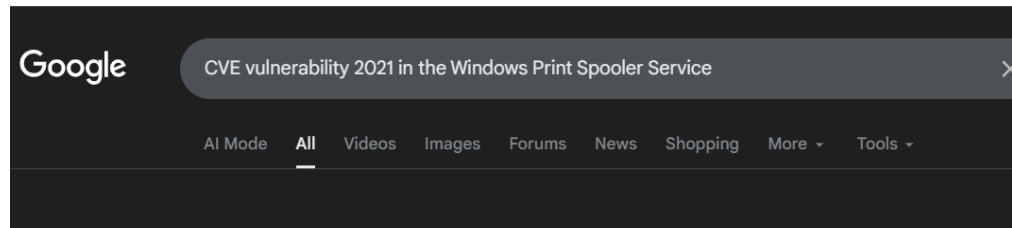
III. Tool Used

- **Web Browser / Search Engine** – Used to research vulnerability information

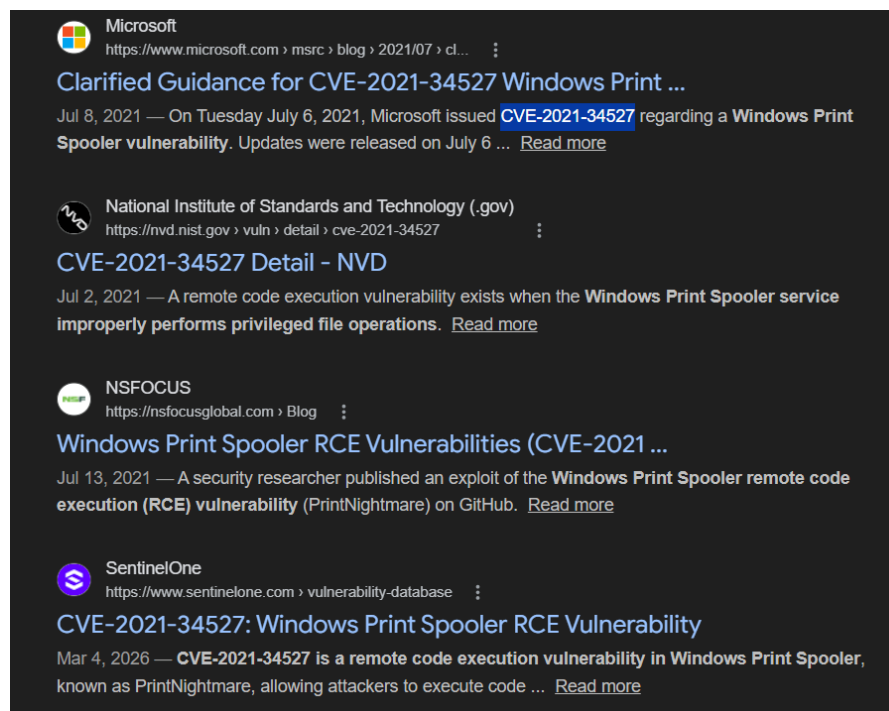
IV. Methodology



1. Searched for information about:
Windows Print Spooler vulnerability 2021



2. Found a well-known vulnerability associated with it



3. Identified the CVE ID:
CVE-2021-34527
4. Formatted it into the required flag format:
picoCTF{CVE-2021-34527}
5. Submitted the flag successfully

V. Result

picoCTF{CVE-2021-34527}



VI. Explanation

What is a CVE?

CVE stands for:

Common Vulnerabilities and Exposures

Think of it like:

A CVE is like a **name tag for a computer bug** 🐞

For example,

CVE-2021-34527

Means:

- **2021** → year discovered
- **34527** → unique ID number

Think of it like:

A **student ID number**, but for security problems

What is a Print Spooler?

The **Print Spooler** is a service in Windows that:

Manages printing jobs

Simple analogy:

- You click **Print** 🖨️
- The spooler puts your file in a **waiting line**
- Then sends it to the printer

Think of it like:

A **queue in a cafeteria line**

What was the vulnerability?

The vulnerability allowed something dangerous:

What is RCE?

RCE = **Remote Code Execution**



Simple meaning:

Someone from far away can run code on your computer

Real-life analogy:

Imagine:

- You own a computer 
- A stranger from the internet presses buttons on it

That's RCE

What happened in this case?

The Print Spooler had a bug that allowed attackers to:

- send malicious instructions
- trick the system
- execute code remotely

This vulnerability became famous as:

CVE-2021-34527

(also known as **PrintNightmare**)

VII. Conclusion

This challenge highlights the importance of understanding real-world vulnerabilities and how they are documented.

By researching the Windows Print Spooler vulnerability, the correct CVE identifier was found and used to retrieve the flag.

This reinforces a key concept in cybersecurity:

Knowing how to research and identify vulnerabilities is just as important as exploiting them.

— The Analyst: Hyposelenia